

Laboratory work 5

Two-dimensional arrays

Objective: learn to work with two-dimensional arrays and program them in the C programming language.

Short theoretical information

5.1. The concept of a two-dimensional array.

A two-dimensional data array is a table consisting of several rows. Elements of a two-dimensional array can be displayed in table view:

a_{00}	a_{01}	...	a_{0m}
a_{10}	a_{11}	...	a_{1m}
...	...	...	...
a_{n0}	a_{n1}	...	a_{nm}

Fig. 5.1

The general format of the design of the description of a multidimensional array (N - dimensional) is as follows:

<Type> <Array Name> [<p1>] [<p2>] ... [<pN>]

where p1, p2, pN set the dimensions for each dimension.

The number of dimensions determines the dimensionality of the array: two-dimensional arrays have two indices, three-dimensional arrays have three, etc.

Elements of a two-dimensional array are defined by the name of the array and two indices (for two-dimensional arrays): the first index determines the number of the row, the second - the number of the column, at the intersection of which the element is placed.

The call to the scalar element of the array is carried out: $a[i][j]$, where i, j are indices.

Consider examples of the description of two-dimensional arrays.

int a[3][4];

float b[10][10];

In the first example, a 3 x 4 matrix will be created, i.e. 3 rows and 4 columns whose elements are integer values. The second contains a 10 x 10 matrix with real numbers. The elements of both arrays will be zeros, since the arrays are uninitialized.

Initialization of arrays. In the C language, arrays can be initialized during description. Initialization of the array - acquisition of defined values by the elements of the array.

Initialization of a two-dimensional array of 3 x 5 numbers from 1 to 15 can look like:

const int n=3;

const int m=5;

```
int ary[n][m] = {  
    { 1, 2, 3, 4, 5 },  
    { 2, 4, 6, 8, 10 },  
    { 3, 6, 9, 12, 15 }  
};
```

5.2. Processing of two-dimensional array elements

Input-output and processing of two-dimensional arrays is performed using nested loops. As a rule, the size of arrays determines the number of cycles. Example:

```
for (int i = 0; i < 6; i++) {  
    for (int j = 0; j < 8; j++) {  
        ary[i][j] = (i + 1) * 5 + (j + 1);  
    }  
}
```

Each element of the array will contain a value determined by the formula $(i + 1) * 5 + (j + 1)$, where i, j are the row and column, respectively.

Actions on array elements are performed in the same way as on simple variables of the corresponding types.

To enter array elements from the keyboard and output array elements to a line, it is necessary to organize a nested loop and enter all values element by element.

To fill the array with pseudo-random numbers and display the elements of the array in the form of a matrix, it is necessary to organize a loop and initialize the generator of pseudo-random numbers using the `srand()` function. The generator must be initialized only once - at the beginning of the program. Using the function `rand()%N+1`, generate for each element of the array a number from the range 1...N.

Example 5.1.

Write a program that generates a two-dimensional array of size 10 by 10 and displays it on the screen.

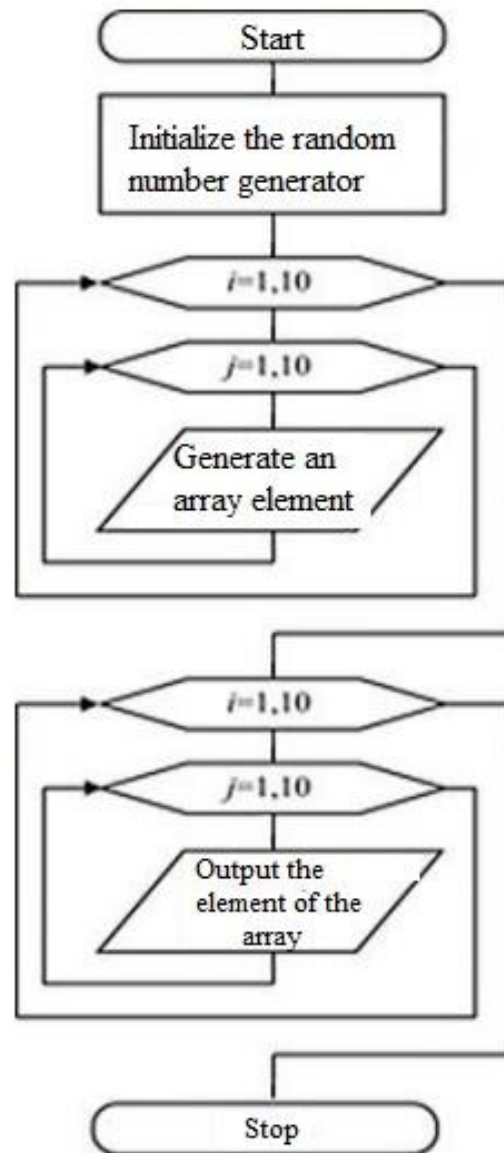


Fig. 5.1 - Block diagram of the algorithm of the program

Program text

```

#include "stdio.h"
#include <stdlib.h>
#include <time.h>
int main()
{
  int a[10][10];
  int i, j;
  srand(time(0));
  for (i = 0; i < 10; i++)
  for (j = 0; j < 10; j++)
  a[i][j] = rand() % 100;
  for (i = 0; i < 10; i++)
  {
    printf("\n");
  }
}

```

```

for (j = 0; j < 10; j++)
printf("%3d", a[i][j]);
}
return 0;
}

```

Results of the program

```

96 22 3 85 63 75 44 0 12 2
41 74 77 55 44 28 55 74 84 63
23 12 49 65 2 69 47 86 89 99
3 22 33 15 8 13 14 24 40 53
2 7 85 66 95 74 11 21 18 0
4 10 78 52 36 38 50 24 19 52
18 18 43 86 81 70 2 29 87 67
99 55 10 12 45 63 89 96 98 5
81 89 7 33 44 12 1 8 78 34
66 96 55 68 77 91 18 36 88 74

```

Example 5.2.

Write a program that will convert a given two-dimensional array into a one-dimensional one using nested loops.

Verbal solution algorithm:

1. Declare a two-dimensional array a and a variable k.
2. Declare a one-dimensional array b with dimensions $m \times n$, where m is the number of elements of array a in a row, n is the number of elements of array a in a column;
3. Fill array a with numbers entered from the keyboard.
4. Set the counter value k equal to zero: $k = 0$;
5. With the help of two nested loops, overwrite the two-dimensional array a into the one-dimensional array b: $k = k + 1$; $b[k] = [i, j]$;
6. With the help of a loop, output the resulting array of

Program text

```

#include "stdio.h"
#include <stdlib.h>
#include <locale.h>
int main()
{
setlocale(LC_ALL, "Ukrainian");
int a[10][10];
int b[100];
int i, j, k;
for (i = 0; i < 10; i++)
for (j = 0; j < 10; j++)
a[i][j] = rand() % 10;
printf("Input data:\na={");
for (i = 0; i < 10; i++)

```

```

{
printf("\n");
for (j = 0; j < 10; j++)
printf("%2d", a[i][j]);
}
printf("}\nOutput data:
b={");
k = 0;
for (i = 0; i < 10; i++)
for (j = 0; j < 10; j++)
{
b[k] = a[i][j];
k++;
}
for (i = 0; i < 100; i++)
printf("%2d", b[i]);
printf("}\n")
return 0;
}

```

The result of the program

Incoming data:

```

a={
5 3 4 5 6 3 4 4 8 1
7 2 1 2 8 4 9 9 6 4
6 3 4 1 7 7 2 7 7 6
5 5 9 1 2 7 4 6 1 2
7 8 4 0 3 4 4 2 3 2
8 2 5 2 3 4 2 7 2 9
4 9 4 5 0 8 3 4 4 6
9 3 7 5 3 0 0 9 4 5
2 2 7 9 5 2 4 7 0 8 }

```

```

Output data: b={5 3 4 5 6 3 4 4 8 1 7 2 1 2 8 4 9 9 6 4 6 3 4 1 7 7 2 7 7 6 5 5 9 1 2
7 4 6 1 2 7 8 4 0 3 4 4 2 3 2 8 2 5 2 3 4 2 7 2 9 4 9 4 5 0 8 3 4 4 6 9 3 7 5 3 0 0 9 4
5 2 2 7 9 5 2 4 7 0 8 }

```

Tasks of laboratory work

Task 1

1. Given matrix A of dimension 6×3 . Select an element $A[i, j]$ arbitrarily and delete the column in which it is located.
2. Given matrix A of dimension 3×6 . Turn relative to the main diagonal.
3. Given matrix A of dimension 10×8 . Turn 180° to the left.
4. Given matrix A of dimension 10×4 . Rearrange the two given lines.

Task 2

1. Given matrix A of dimension 7×9 . Find the arithmetic mean of its elements and the number of elements smaller than the arithmetic mean.
2. Given matrix A of dimension 10×4 . Rearrange the two given rows and sort the matrix.
3. Given matrix A of dimension 3×7 . Find the sum of the elements of each line. Write the result in a one-dimensional array and sort it.
4. Given a matrix A of dimensions 15×10 . Find the row with the smallest element and swap it with the last one.

Task 3.

1. Given matrix A of dimension 7×9 and matrix B of dimension 5×4 . Perform $A \times B$ multiplication, record the result in the C matrix.
2. Given square matrices A and B of dimensions 5×5 . Perform a direct sum of square matrices $A \oplus B$, write the result in matrix C.
3. Given matrix A of dimension 7×9 and matrix B of dimension 5×4 . Add $A+B$, record the result in matrix C.
4. The given matrix A of dimensions 7×9 Perform its transposition..